

Capstone Project Document

AI-powered Film Photography Contest Management Platform	
Group Members:	Tran Phat Tai - 086206011804 Tran Quy Minh Thien - 079207025398 Vo Quoc Khanh - 066207017208 Nguyen Thi My Uyen - 068307002174 Huynh Nguyen Duy Tan - 054207001780
Lecturer:	Nguyen Van Chien

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Finally, although we have worked hard to ensure accuracy, errors may still exist. We welcome any feedback and suggestions from our lecturers to further improve this project.

Definition and Acronyms

Acronym	Definition	Description
API	Application Programming Interface	A set of rules and protocols that allows different software applications to communicate with each other.
BA	Business Analysis	The practice of enabling change in an enterprise by defining needs and recommending solutions that deliver value to stakeholders.
BR	Business Rule	A rule that defines or constrains some aspect of business and always resolves to either true or false.
ERD	Entity Relationship Diagram	A flowchart that illustrates how "entities" such as people, objects, or concepts relate to each other within a system.

Acronym	Definition	Description
EXIF	Exchangeable Image File Format	A standard format for specifying meta-data tags embedded in digital image files (e.g., Camera Model, Lens, Film Stock/ISO).
GUI	Graphical User Interface	A type of user interface that allows users to interact with electronic devices through graphical icons and visual indicators.
MHs	Man-Hours	An estimated unit of work equal to one hour of continuous labor by one project team member.
pHash	Perceptual Hashing	A fingerprinting algorithm used to analyze image contents and generate a unique hash for duplicate image detection.
PM	Project Manager	The person responsible for planning, executing, monitoring, controlling, and closing projects.
RBAC	Role-Based Access Control	A security approach that restricts system access based on defined user roles (e.g., Candidate, Judge, Organizer).
SDD	Software Design Description	A document that describes the design of a software system, including its architecture, components, and interfaces.
SPMP	Software Project Management Plan	A document that defines the project plan, specifying the tasks, schedule, resources, and management approach.
SRS	Software Requirement Specification	A document that describes what the software will do and how it will be expected to perform.
UAT	User Acceptance Test	The final phase of the software testing process where actual software users test the software to make sure it can handle required tasks in real-world scenarios.
UC	Use Case	A list of actions or event steps typically defining the interactions between a role and a system to achieve a goal.

I. Project Introduction

1 Overview

1.1 Project Information

Project Name: AI-powered Film Photography Contest Management Platform

Software Type: Web Application

Purpose: To streamline the organization, submission, judging, and exhibition processes of film photography competitions.

Group Name: Group 7

2 Project Team

Full Name	Role	Email
Nguyen Van Chien	Lecturer	chiennv@ut.edu.vn
Tran Phat Tai	Leader	taitp1804@ut.edu.vn
Tran Quy Minh Thien	Member	thientqm795398@ut.edu.vn
Vo Quoc Khanh	Member	khanhvq667208@ut.edu.vn
Nguyen Thi My Uyen	Member	uyenntm682174@ut.edu.vn
Huynh Nguyen Duy Tan	Member	tanhnd541780@ut.edu.vn

3 Product Background

Film photography has become increasingly popular in recent years, leading to a growing number of photography contests. However, most contests are still managed using separate tools such as online forms, spreadsheets, emails, and cloud storage, resulting in inefficient workflows and difficulties in submission verification, judging, and data management.

Unlike digital photography contests, film photography requires additional technical metadata such as film stock, camera, lens, ISO, developing laboratory, and scanning information. Existing contest management systems generally do not support these specialized requirements.

To address these limitations, this project proposes an AI-powered Film Photography Contest Management Platform. The platform centralizes contest management, supports film-specific metadata, and integrates AI for duplicate image detection, AI-generated image detection, automatic image categorization, and contest analytics, improving efficiency, transparency, and fairness throughout the competition process.

4 Existing Systems

4.1 Picter

Link: <https://www.picter.com>

Description: A professional platform for managing photography competitions, open calls, grants, and awards. It is widely used by organizations such as World Press Photo, Leica, and Magnum.

Actors: Organizers, judges, photographers, administrators.

Key Features:

- Online contest creation and management.
- Photo submission and portfolio management.
- Online judging and scoring.
- Result publication and digital archive.

Pros: Complete end-to-end contest management; professional judging workflow; supports large-scale international competitions.

Cons: Not designed specifically for film photography; does not provide film-specific metadata management (film stock, camera, developing lab, etc.); limited AI verification features.

4.2 CameraComp

Link: <https://www.cameracomp.com>

Description: A cloud-based competition management system developed for photography clubs and federations.

Actors: Club administrators, judges, photographers.

Key Features:

- Competition management.
- Image submission.
- Online and offline judging.
- Member and event management.

Pros: Simple and intuitive interface; integrated competition and member management; supports multiple judging methods.

Cons: Focuses on photography clubs rather than film photography contests; no AI-generated image detection; no film metadata management.

Compared with the systems above, the proposed platform is specifically designed for film photography contests. It focuses on film-specific metadata, AI-based verification, and a more specialized workflow for organizers, judges, and participants.

5 Business Opportunity

The growing popularity of film photography has increased the demand for efficient contest management. However, most organizers still use separate tools, making submission management, judging, and verification inefficient. This project provides a centralized platform with AI support to improve efficiency, transparency, and fairness for photography contests.

6 Software Product Vision

The project aims to develop a web and mobile platform for managing film photography contests. It supports the complete contest workflow, from registration and submission to judging and result publication, while integrating AI for duplicate image detection, AI-generated image detection, and automated image categorization to enhance contest management.

7 Project Scope & Limitations

7.1 Major Features

The following primary functional modules are included:

- **FE1 – User & Role Management:** Provides authentication, role-based access control (RBAC), and profile management for Candidates, Judges, and Organizers/Admins.
- **FE2 – Contest Management:** Enables Organizers to create and configure new contests, set submission rules, manage timelines, oversee participant lists, and assign judges.
- **FE3 – Submission System:** Allows candidates to upload high-resolution film scan images, input detailed technical specifications (EXIF metadata, Camera, Lens, Film Stock/ISO, Aperture, Shutter Speed), and track submission statuses.
- **FE4 – Judging & Evaluation:** Provides a dedicated judging panel featuring high-resolution image viewing tools (Zoom/Pan), multi-criteria evaluation scoring, and feedback/commenting capabilities.

- **FE5 – Automated Processing & Anti-cheat:** Automatically extracts EXIF data from uploaded images and utilizes perceptual hashing (pHash) with Hamming distance calculations to detect and flag potential image plagiarism or duplicate submissions.
- **FE6 – Exhibition Gallery:** A public online gallery showcasing winning or selected submissions, equipped with advanced filtering options by Film Stock, Camera model, or Contest.

7.2 Limitations & Exclusions

The following points detail the operational constraints and out-of-scope boundaries:

- **LI-01:** File size and format restrictions: Supports only .jpg, .jpeg, and .png formats, with file sizes strictly limited between 2 MB and 25 MB.
- **LI-02:** Automated metadata extraction depends entirely on intact EXIF data within uploaded film scan files; stripped EXIF files require manual input and administrative flagging.
- **LI-03:** Perceptual hashing (pHash) and Hamming distance duplication detection are processed asynchronously via message queues, resulting in slight processing delays rather than real-time feedback upon upload.
- **LI-04:** No support for native mobile applications (iOS/Android); the system is strictly developed as a responsive web application.
- **LI-05:** No integration with online payment gateways or photo editing/post-processing tools (candidates must upload finalized image files, and all financial transactions are out of scope).

II. Project Management Plan

1 Overview

1.1 Scope & Estimation

WBS	Item	Complexity	Est. Effort (MHs)
1	FE01. User & Role Management		40
1.1	Authentication System (Login, Register, OAuth Google, JWT)	Simple	10
1.2	Password Recovery & Email Verification	Simple	8
1.3	User Profile & Portfolio Management	Medium	10
1.4	Role-Based Access Control (RBAC) & User Management (Admin)	Medium	12
2	FE02. Contest Management		48
2.1	Contest Creation & Basic Setup (Rules, Prizes, Categories)	Medium	14
2.2	Timeline & Phase Transition Configuration (Auto Scheduler)	Medium	12
2.3	Jury Assignment & Judging Criteria Setup	Medium	12
2.4	Contest Status Management & Moderator Dashboard	Simple	10
3	FE03. Submission System		64
3.1	Presigned Cloud Storage Upload (S3/Cloudinary) for Hi-Res Scan	Complex	20
3.2	Film Specifications & Metadata Input Form (Film Stock, Camera, Lens)	Medium	14
3.3	Negative Film / Contact Sheet Proof Attachment Upload	Medium	16
3.4	Submission Drafts & Candidate Entry History	Simple	14
4	FE04. Judging & Evaluation		56
4.1	Blind Judging Portal (Anonymized Candidate Data)	Medium	12
4.2	High-Resolution Image Viewer (Zoom/Pan HD & Negative Inspector)	Medium	16
4.3	Multi-Criteria Scoring System & Feedback Panel	Complex	16
4.4	Final Leaderboard Calculation & Winner Approval	Medium	12
5	FE05. Automated Processing & Anti-cheat (AI Subsystem)		72
5.1	Auto EXIF Metadata Extraction Engine	Medium	18

WBS	Item	Complexity	Est. Effort (MHs)
5.2	EXIF vs User Declaration Mismatch Verification	Medium	14
5.3	pHash Duplication Detection & Fingerprinting Algorithm	Complex	24
5.4	Fraud Flagging System & Admin Inspection Dashboard	Medium	16
6	FE06. Exhibition Gallery & Community		32
6.1	Public Gallery & Masonry Display Grid	Simple	8
6.2	Advanced Filtering (Film Stock, Camera Model, Contest, Year)	Medium	10
6.3	Public Voting & Anti-Spam Community Engagement	Medium	8
6.4	Social Sharing & Certificate Generation for Winners	Simple	6
7	FE07. System Architecture, QA & DevOps		88
7.1	SRS / SDD Documentation & System Architecture Design	Medium	24
7.2	Database Schema Design (ERD) & Base API Setup	Medium	16
7.3	QA Test Plan, Test Cases Execution & Bug Fixing	Medium	30
7.4	Docker Containerization, CI/CD Pipeline & Deployment	Medium	18
TOTAL			400 MHs

1.2 Project Objectives

Testing Stage	Test Coverage	No. of Defects	% of Defect	Notes
Reviewing	100% artifacts reviewed (SRS, SDD, ERD, Wireframes)	-	-	Peer review and supervisor feedback
Unit Test	$\geq 70\%$ code coverage for Core BE and AI modules	0 critical; ≤ 10 minor	$\leq 10\%$	Automated testing (JUnit, Jest, PyTest for AI functions)
Integration Test	$\geq 85\%$ API and async pipeline coverage	0 critical; ≤ 8 minor	$\leq 8\%$	Postman / Swagger and S3 upload / AI queue testing
System Test	$\geq 95\%$ major features (Contest, Submission, AI Inspection, Judging)	0 critical; ≤ 5 minor	$\leq 5\%$	Staging environment regression and performance testing for high-resolution uploads
Acceptance Test (UAT)	100% critical user flows and AI verification accuracy	0 critical / blocker	0%	UAT with stakeholders (organizers, jury, candidates)

1.3 Project Risks

No.	Risk Description	Impact	Possibility	Response Plans
1	Changing requirements from stakeholders (organizers, jury)	Medium	Medium	Baseline SRS plus change control process; weekly review and sign-off with supervisor
2	Limited domain knowledge in film photography (film stocks, EXIF structures, scan types) and AI anti-cheat models	High	Medium	Consult film photography experts/advisors; early PoC for EXIF extraction and pHash algorithm; prototype feedback
3	Third-party integrations and services (AWS S3 / Cloudinary for high-resolution image storage, Docker, Redis, Resend / SMTP) are unstable or exceed free-tier limits	Medium	Medium	Early PoC in sandbox; image compression and thumbnail generation; retries and fallback mechanisms; service cost monitoring
4	Data privacy and security (protecting unpublished contest entries and candidate PII)	High	Low	Implement RBAC, anonymized submission pipeline for blind judging, S3 presigned URLs with expiration, and data encryption at rest and in transit
5	Resource availability or absence among team members	Medium	Medium	Cross-training tech stacks (front-end / back-end / AI); backup feature ownership; daily standups; sprint buffer time
6	Performance degradation under peak loads (simultaneous high-resolution image uploads near contest deadline)	High	Medium	Direct-to-S3 presigned upload (bypass core server); Redis caching; asynchronous background AI processing queue; rate limiting and load testing in Sprint 6

2 Management Approach

2.1 Project Process

Methodology: The project applies the Agile Scrum methodology.

Iteration Model: 1-week sprints, total of 7 sprints (7 weeks).

Sprint Breakdown:

- **Sprint 1:** Project initiation, requirement elicitation (SRS), architecture & DB design (SDD), UI/UX wireframing, and AI Proof of Concept (EXIF & pHash).
- **Sprint 2:** Development of FE01 (User Authentication, Profile Management, & RBAC).
- **Sprint 3:** Development of FE02 (Contest Creation, Timeline, & Jury Assignment).
- **Sprint 4:** Development of FE03 (Submission System with Presigned S3 Upload & Film Specifications).
- **Sprint 5:** Development of FE04 (Blind Judging Portal, HD Image Viewer, & Scoring System).
- **Sprint 6:** Development of FE05 (AI Processing Pipeline: Auto EXIF & pHash Duplicate Detection) & FE06 (Exhibition Gallery).
- **Sprint 7:** FE07 Integration, System Testing, Bug Fixing, UAT, Docker Containerization, & Cloud Deployment.

Scrum Ceremonies: Daily Stand-up (15 minutes), Weekly Sprint Planning, Sprint Review, and Sprint Retrospective.

Artifacts: Product Backlog, Sprint Backlog, Burndown Chart, Weekly Working Increment.

2.2 Quality Management

The quality objective is to ensure a stable, accurate, secure, and performant platform for managing film photography contests and automated image inspection within the 7-week timeframe.

The approach includes:

- **Defect Prevention:**
 - Enforce clean code conventions and modular architecture (Separation of Core API and AI Services).
 - Database normalization (PostgreSQL), constraints, and indexing for fast query performance on media records.
- **Reviewing:**

- Mandatory Code Review with at least 1 Peer Reviewer and Leader approval before merging Pull Requests.
- Peer review for technical documentation (SRS, SDD, ERD, Test Cases).

- **Unit Testing:**

- **Tools:** Jest/React Testing Library (Frontend), JUnit/PyTest (Core BE & AI Engine).
- **Goal:** $\geq 70\%$ code coverage, focusing on critical API logic, EXIF parsers, and hashing algorithm utilities.

- **Integration Testing:**

- **Tools:** Postman / Newman for REST API workflows.
- **Scope:** Asynchronous task queue handling (Redis/Celery) for background image processing and Direct-to-Cloud (AWS S3 / Cloudinary) presigned upload pipelines.

- **System Testing:**

- Conducted in Week 7 on a Staging Environment hosted on AWS / Cloud VPS.
- Full regression testing of all modules (FE01 to FE07).
- Performance and Load Testing (using Apache JMeter or K6) for peak submission scenarios near contest deadlines and Hi-Res image viewing.

- **Acceptance Testing (UAT):**

- User Acceptance Testing (Week 7) with primary stakeholders: Contest Organizers, Jury Members, and Candidates/Photographers.
- Ensure 100% of critical flows (Contest creation, Entry submission, AI flagging, and Blind judging) pass successfully before final presentation.

2.3 Training Plan

Training Area	Participants	When, Duration	Waiver Criteria
Next.js / ReactJS (Frontend)	Uyen, Tan	Week 1, 2 days	≥ 1 year React experience
Node.js / Java Spring (Backend)	Khanh, Tai	Week 1, 2 days	≥ 1 year Backend experience

Python & OpenCV / pHash / EXIF Parsing	Thien	Week 1, 2 days	Prior AI / Image Processing experience
PostgreSQL Schema & Media Indexing	All	Week 1, 1 day	Prior DB design experience
Git/GitHub Workflow (PR, Code Review)	All	Week 1, 0.5 day	Familiar with Git flow
AWS S3 / Cloudinary Presigned Direct Upload	Khanh, Uyen, Tan	Week 2, 1 day	Prior Cloud Storage experience
Security & RBAC Authorization (JWT)	All	Week 2, 0.5 day	Completed Security course
Redis Queue & Async Task Handling	Khanh, Thien	Week 3, 1 day	Prior Message Queue experience
Deep Zoom HD Image Viewer (OpenSeadragon)	Uyen, Tan	Week 4, 0.5 day	Prior Canvas/Image lib experience
Docker Containerization & Cloud Deployment	Tai, Khanh	Week 6, 1 day	Prior DevOps/Docker experience

3 Project Deliverables

No.	Deliverable	Due Date	Notes
1	Project Charter, SRS, SDD & AI PoC Report	Sprint 1 (Week 1)	
2	Authentication & User Management Module (FE01)	Sprint 2 (Week 2)	
3	Contest Management & Jury Assignment Module (FE02)	Sprint 3 (Week 3)	
4	Submission System & Direct Cloud Storage (FE03)	Sprint 4 (Week 4)	
5	Blind Judging Portal & HD Scoring System (FE04)	Sprint 5 (Week 5)	
6	AI Pipeline Integration & Exhibition Gallery (FE05 & FE06)	Sprint 6 (Week 6)	
7	Integration Test, Docker Package & Cloud Deployment (FE07)	Sprint 7 (Week 7)	

4 Responsibility Assignments

Responsibility	Tai	Thien	Khanh	Uyen	Tan
Project Planning & Tracking	D	R	S	R	I
Prepare Project Charter & SRS Document	D	R	S	R	S
Architecture & DB Design (SDD, ERD)	S	R	D	I	I
UI/UX Wireframing & Design (Figma)	R	I	I	D	D
Core Backend API Implementation (Spring Boot)	S	I	D	I	I
Direct Cloud Storage & Presigned URL Integration	I	S	D	I	R
AI Pipeline Implementation (EXIF & pHash Engine)	R	D	S	I	I
Frontend Implementation - Candidate & Gallery (FE03, FE06)	I	I	I	S	D
Frontend Implementation - Jury Portal (FE04)	I	I	I	D	S
System Integration & Testing (Testing & QA)	R	R	R	R	D
Final Deployment & Project Documentation	D	R	R	R	R

5 Project Communications

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Weekly Planning Meeting	All team members	Plan tasks for upcoming sprint, review SRS/SDD, and assign roles	Every Friday (1 session/week)	Online meeting (Google Meet)
Weekly Progress Review	All team members	Share development progress, adjust timeline, discuss blockers	Every Wednesday (1 session/week)	Google Docs, Zalo group

Internal Feature Demo	Relevant developers + Reviewer (Leader)	Validate completed modules (Spring Boot, Python AI, React UI) before merging	After each feature completion	Live screen share via Google Meet or Zalo video
Documentation Review	Assigned members (Tai, Khanh)	Review and update SRS, SDD, ERD, and User Manual documents	Every week	Google Docs
Code Review Session	Feature developer + Reviewer	Review code logic, API endpoints, pHash/EXIF algorithms, and coding conventions	Before merging Pull Request to main branch	GitHub Pull Request
Urgent Bug or Blocker Sync	Involved members only	Quickly resolve critical technical issues, API mismatch, or Cloud upload errors	Ad-hoc (as needed)	Zalo Chat / Call
Shared Asset Management	All members	Store and collaborate on design files (Figma), diagrams (Draw.io), docs, and slide presentations	Continuous	Google Drive / Figma / GitHub

6 Configuration Management & Infrastructure

Category	Tools / Infrastructure
Frontend	React / Next.js
Backend	Java Spring Boot, Python FastAPI (AI Subsystem)
Database & Cache	PostgreSQL
Cloud Storage	AWS S3 / Cloudinary (Presigned / Signed URL)
IDEs / Editors	VS Code, PyCharm
Diagramming & UI	Figma, Draw.io
Documentation & VCS	Google Docs, GitHub
Testing & API Spec	Postman, Swagger / OpenAPI